

# Elephant V1.0 Writeup

*Ancient Proverb: "How do you eat an elephant? One bite at a time."*

**Goal:** Create a productivity/to-do list that minimizes the amount of information presented to the user for their "current task."

**Scenario:** let's say a user has 3 projects with 10 items each, and an additional errands list of 10 items. If they see the entire list all at once, they will feel overwhelmed. Instead, our app shows them 1 item at a time. Meanwhile, the "Master List" of all the items is "mixed" so that the User doesn't get shown all 10 items of 1 project and then the next 10 items of the next project. Instead, they get shown a healthy mixture of 1 item from each Project and an Errand every 2-3 Items, to give them something easy to accomplish in between the hard stuff.

## App Objects

**Items:** We call them Items rather than tasks because "tasks" sound like a chore. "Items" seems like a building block. We don't want the user to feel like they are doing a never-ending list of chores, but rather building a foundation for a better future.

**Errands:** These are project-less Items. They are for the little things that come up during the day. These are the "snacks" that the user can enjoy, between the meatier "Items" that are higher pressure and take more time.

**Projects:** Projects are lists of Items in sequential order. The user assigns a Priority Score to each Project. Priority 5 = super important. Priority 1 = less important. The higher the priority, the more often the Project's Items will appear in the Master List.

## App Lists

**Master List:** A hidden list that the user never sees (unless they absolutely want to). This list is the master "To-Do" list which includes all the Items (both Project Items and Errands).

This Master List is like a line at the doctor's office. Once given an order, Items do not lose their place in line (except when modified by a function).

**Completed List:** An archived list of completed Items, all tagged with the timecode of completion when the user pressed "Completed!"

**Active Project List:** A list of ongoing projects.

**Inactive Project List:** A list of upcoming projects (all Items are inactive).

**Completed Project List:** A list of completed projects.

**Project Items List:** A list of Items within a Project

## App Pages

**Do Now Page:** This is a super minimal page that shows only the Item or Errand at the top of the list. It also shows in smaller letters, the Project that the Item is a part of. If it's an Errand, the Project text field is blank.

**Home Page:** This is the homepage of the app, which is the first page the user sees when opening. From this page, they will be able to navigate to the Do Now Page, Completed List Page, Projects List Page, or Settings Page.

**Master List Page:** Presents the Master List of all Items, sorted by "Date Added"

**Completed List Page:** Presents the Completed List of Items and Projects (1 tab for all items, and 1 tab for projects).

**Project List Page:** Presents the Project Lists (2 tabs total: Active Projects & Inactive Projects). Can be sorted by either Alphabetical order or Priority.

**Edit Project Page:** A working page which allows the user to modify their project.

## **Page Objects & Buttons**

### **Do Now Page Display**

**Text 1:** The #1 Item from the top of the Master List.

**Placement:** This text should be presented in the center, simply & prominently, with a frame around it (stylized, flowery gold frame).

**Text 2:** The Project name of the #1 Item (blank if the #1 Item is an Errand).

**Placement:** This text should be 35-50% smaller than Text 1, and should appear directly underneath Text 1 (outside of the gold frame).

**Button 1:** "Completed!" When pressed, this button triggers the functions CompleteItem and ReprocessList.

**Placement:** This button should be green and placed in the lower right corner.

**Button 2:** "Take a Bite" When pressed, this button triggers the functions RenameItem and CreateNewItem

**Placement:** This button should be red and placed in the lower left corner.

**Button 3:** Add Item button, which triggers the function CreateNewItem

**Placement:** This is a simple "+" sign on the top right hand corner of the page

**Button 4:** Back to Homepage. Upper left hand corner (self-explanatory).

### **Homepage Text & Buttons**

On the Home Page, the user will be able to see a welcoming text and profile picture "Welcome, Lionel!"

Below the profile, there will be 4 buttons in this order:

**Button 1:** Start Working (transfer to Do Now Page)

**Button 2:** Manage Projects (transfer to Project List Page)

**Button 3:** Completed List (transfer to Completed List Page)

**Button 4:** Settings (transfer to Settings Page)

**Button 5 (hidden):** View Master List (transfer to Master List Page). This button is hidden by default and can be activated by toggling the setting "Show Master List Button on Homepage."

Finally, on the top right hand corner, there is also an "Add Item" button similar to the Do Now Page, which triggers the function CreateNewItem

### **Project List Page Display**

This page has 2 tabs, with the first tab open by default.

**Tab 1 Header:** Active Projects

Presents a table of the active projects (from the Active Project List), with the following information:

Project Name | Priority | Optional Columns: Create Date, Due Date (blank if none)

Tab 2: Header - "Inactive Projects". Presents the same information as Tab 1 but from the Inactive Project List

Table Dynamics:

The projects table should be scrollable similar to how one scrolls through a text selector on an iPhone.

When clicking on the project, the user will be taken to the "Edit Project Page"

There is also a button called "Add Project" which shows the following form:

1) Project Name

2) Project Priority (select 1-5)

This creates a blank project which can then be edited via the Edit Project Page

### Edit Project Page Display

This page is opened from the Project List page (whether by clicking an existing Project or by creating a new one) and shows the attributes of the project which the user clicked on.

Text 1: Project Name (Clicking on it allows us to edit the name)

Text 2: Project Priority (Clicking on this allows us to select the Priority, numbers 1->5)

Text 3: Created Date (Not editable)

Text 4: Due Date (Editable, date selector popup)

Below these attribute texts, we have a list of Items that are within this project (in sequential order).

If the user clicks on the item, they can edit the item name. If they swipe left, they can delete it. They can long-press and hold to drag the item up and down the list in priority.

Button 1: Add Item To Project (at the bottom of screen)

This will add an Item to the project (if another Item is selected, it will add the New Item below that Item, if no Item is selected, it will add the New Item to the bottom of the list).

### Master List Display

This is a hidden page, but still needs to be designed because it displays the Master List.

Text: Master List (View with caution)

It should show the list of all Active Items, in the order that they were added to the Master List, as well as their projects (in a separate column on the right hand side).

### Master List Functionality

The functionalities described here is the most important part of our app, even if its processes are mostly hidden from the user.

The list starts off with 0 items. When a user creates an Item without a project (Errand), the Item is added to the list in 1st position. Subsequent Items get added to the end of the list.

There is no functionality to "drag" or "reorder" items on this list, like there is on the Edit Project Page.

When a user creates an Item with a Project, the Item is first added to the Project's "Inactive Items" list. It is not yet added to the Master List (because it is tagged as Inactive).

When an Item is "Completed", it triggers the "CompleteItem" and "ReprocessList" function. The CompleteItem function does the following:

- 1) Takes out the Index 1 Item from the Master List and puts it into the Completed Items list.
- 2) Displays the next Item and the Item attributes (the current index 1 Item and its project text if applicable)
- 3) Updates the index of every single Item of the Master List (for example the 5th Item is now #4)

Next, it triggers the ReprocessList function.

- 4) Pings each Project to see if they are "ready." This triggers the following calculation for each Project:

PingProjectForItem (PPFI)

Variables needed for this function:

1. Position of the Last Active Project Item in the Master List (PLPI)
2. Total Number of Master List Items (TNML)

Calculated variables needed for this function:

Project Insertion Threshold (PIT):  $1/\text{Project Priority}$

Project Ready Score (PRS):  $1-(\text{PLPI}/\text{TNML})$

If  $\text{PRS} > \text{PIT}$ , activate the topmost inactive item of the Project. Otherwise, do nothing for this project.

When this item is "activated", it goes to the bottom of the Master List.

Example: There are 10 Master List Items. Project X, with priority 3, has 5 Items, A, B, C, D, E. A and B are active on the Master List at index positions 5 and 8. C, D, and E are inactive and exist only on the Project Items list.

At this point, the "ReprocessList" function is triggered manually by the User.

The PLPI is noted as 8 (position of Item B)

The TNML is noted as 10.

The PIT is calculated for the project:  $1/3 = 0.3333$

The PRS is calculated for the project:  $1-(8/10) = .2$

Since the PRS is less than the PIT, nothing happens.

Next, the User completes Item #1 of the Master List, which triggers the ReprocessList function again.

The PLPI is now 7 (8->7 as all Master List Items move up 1). The TNML is now 9.

The PIT remains the same (0.3333). The PRS is now calculated as:  $1-(7/9) = .222$

Since PRS is still less than the PIT, Project X activates no new Items.

Next, the user adds 2 Errands (Items without projects) to the Master List, which triggers the ReprocessList function again.

The PLPI is still 7 (since no Items were completed, all existing items retained their same index). However, the TNML is now 11. The PIT for Project X is still the same (0.3333).

However, the PRS is now  $1 - (7/11) = .3636$

Now, the PRS is higher than the PIT (.3636 > .3333). Therefore, the "ActivateProjectItem" function is triggered, which now adds the highest Inactive Item (Item C) to the Master List, increasing the number of Master List Items to 12.

Now if we run "ReprocessList" again, we get the following:

PLPI is now 12 (C's position). Project X has 3 Items on the Master List: Items A (Index: 4), B (Index: 7), and C (Index: 12). TNML is also 12. PRS is now  $1 - (12/12) = 0$ . So it is lower than the PIT (.3333) therefore, no new items get activated.

Now let's say 1 item gets completed, and 1 item gets added. ReprocessList runs again, resulting in this:

PLPI is now 11 (C's new position). Project X still has 3 Items on the Master List: A (index 3) B (index 6) and C (index 11). TNML is still 12 (due to 1 Item leaving the list and 1 Item added). PRS is now  $1 - (11/12) = .08333$ . Therefore, no new items are activated.

Regarding Project Items in the Master List: the Master List has 2 versions. The back-end version and the front-facing version.

The back-end version does not add the actual Project Item and its details. Instead it adds a "Project Placeholder" item. So for example, a Master List might look like this:

1. Errand 1
2. Project X Placeholder 1
3. Errand 2
4. Project Y Placeholder 1
5. Project X Placeholder 2
6. Errand 3
7. Project Y Placeholder 2
8. Project X Placeholder 3

The reason for these placeholders is to allow the User to edit the order of Items within their Projects, while the Project Placeholder maintains the Project's "reserved spots" within the Master List.

So to the User, they will see this:

1. Errand 1
2. Project X Item A
3. Errand 2
4. Project Y Item D
5. Project X Item B
6. Errand 3
7. Project Y Item E
8. Project X Item C

However, let's say they decided to add another Item to Project X (Item G) and put it between Items A and B.

So their Project Items list now looks like this:

1. Item A
2. Item G
3. Item B
4. Item C

But the front-end Master List will now look like this:

1. Errand 1
2. Project X Item A
3. Errand 2
4. Project Y Item D
5. Project X Item G
6. Errand 3
7. Project Y Item E
8. Project X Item B

As we can see, the Placeholders allowed Project X to maintain their index positions of 2, 5, 8. But Item C has been taken off the Master List (deactivated). Meanwhile, Item G is now where Item B previously was, and now Item B is now where Item C previously was.

Therefore, the back-end Master List still looks like this:

1. Errand 1
2. Project X Placeholder 1
3. Errand 2
4. Project Y Placeholder 1
5. Project X Placeholder 2 (Adding Item G between items A and B in the Project List, does not affect Project X's placeholder positions)
6. Errand 3
7. Project Y Placeholder 2
8. Project X Placeholder 3

The purpose of placeholders is to maintain continuity of a Project's "spot in line" within the Master List, while still being able to edit individual items within the Project.

### **Do Now Page Functionality**

The Do Now Page has 2 buttons, the "Completed!" green button and the "Take a Bite" red button.

Pressing the Completed button kicks off the CompleteItem and ReprocessList functionalities.

CompleteItem Functionality:

Moves the Index 1 Item from the Master List to the Completed List and fills out the "DateTime Completed" attribute for the Item.

Re-indexes all the items of the Master List (re-calculates the MLP of each item). MLP = Master List Position.

ReprocessList function:

- 1) Re-calculates the Total Number of Master List Items (TNML)
- 2) Pings each project with the function Ping Project For Item (PPFI)
- 3) Re-indexes all items of the Master List (re-calculates the MLP of each item).

Pressing the Take a Bite function executes the following:

- 1) A prompt appears asking the User to edit 2 text fields.

Text field 1: contains the original Item Text, which can be renamed

Text field 2: (blank)

The user can now rename the Text Field 1 and also fill out Text Field 2.

Example: the #1 Item is an Errand which says "Buy 2 cartons of milk from supermarket". However, the User can only find 1 carton of milk at the grocery store. They still need to buy another one later this week, but they don't need to do so right now.

The Take a Bite function gives them a prompt with these 2 text fields:

Text field 1: Buy 2 cartons of milk from supermarket

Text field 2: (blank)

Now, the user can edit the text of both these fields:

Text field 1: Buy 1 carton of milk from supermarket

Text field 2: Buy 1 carton of milk from supermarket

The prompt also includes a "Cancel" and "Complete" Cancel reverts everything back, as if the User did not click on the Take a Bite function. Complete does the following:

- 1) The #1 Item is now renamed to Buy 1 carton of milk from supermarket.
- 2) Now, a new Item (Errand) is created which is added to the bottom of the Master List.

Now, the User can proceed to complete the new renamed Item by pressing the "Completed!" button which kicks off the CompleteItem and ReprocessList functions.

Take a Bite for Project Items

If the #1 Master List Item (MLI) is a Project Item, then the behavior is slightly different. For example, let's look back at the below example, and imagine that Project X has the following details:

Project X: Send Text Message Invites for Sunday Dinner

Item A: Write up the Invite

Item B: Create List of Invitees

Item C: Send invite via text message to all invitees

1. Project X Item A (Project X Placeholder 1) "Write Up the Invite"

2. Errand 1

3. Project Y Item D (Project Y Placeholder 1)

4. Project X Item B (Project X Placeholder 2) "Create List of Invitees"

5. Errand 2

6. Project Y Item E (Project Y Placeholder 2)

7. Project X Item C (Project X Placeholder 3) "Send invite via text message to all invitees"

8. Errand 3

So currently Project X Item A is at the top of the list. The User selects "Take a Bite" and sees the following 2 text fields:

Text field 1: "Write Up the Invite"

Text field 2: (blank)

Let's say the User is having trouble writing up the Invite because they aren't sure what the dinner group wants to eat. They realize they first need to check with their 1-2 friends if they should have a potluck or ordering pizza.

So they will edit the text fields like this:

Text field 1: "Call Paul and Stacy, ask for advice" (rename Item A to this)

Text field 2: "Write Up the Invite based on Paul & Stacy's advice" (create new Item D)

The Project X List ("Send Text Message Invites for Sunday Dinner") now looks like this:

Item A: Call Paul and Stacy, ask for advice

Item D: Write Up the Invite based on Paul & Stacy's advice

Item B: Create List of Invitees

Item C: Send invite via text message to all invitees

Notice how within the project list, the new Item D is now in 2nd place.

So now, what happens to the Master List, is the #1 Item still shows Project X Placeholder 1, but it contains the new text of Item A which is "Call Paul and Stacy, ask for advice". Meanwhile the Project X Placeholder 2 now shows "Write Up the Invite based on Paul & Stacy's advice"

Meanwhile, on the Do Now Page, the User is now looking at the following text:

Item Name: Call Paul and Stacy, ask for advice

Project Name (smaller font): Send Text Message Invites for Sunday Dinner

And they can now call Paul and Stacy, and click "Completed!" once this action is complete.

## Settings Page

In the settings page, there are a few buttons/functions:

Button: Show Master List. Function: allows the User to view the Master List (reveal hidden button on HomePage)

Button: Export Database. This creates a .json file of the entire database (Master List, Completed List, Active Project List, Inactive Project List, and Project Items List for each Project) and gives user to download & save to desktop.



**Button: Import Database.** Imports a .json file from the User's specified location (for example, their desktop). When triggered, this presents the User with a question: "This will overwrite all current data. Are you sure?"

If confirmed, it runs validation checks & overwrites existing Database - this deletes all the existing Database's lists, and repopulates the lists with the imported .json file's lists.

**Button: Export CSV.** Upon clicking this, a prompt will ask them which list they want to export CSV of (Master List, Completed List, Active Project List, etc.)

**Button: Reset App** (deletes all lists, full reset).